



Math 111: Linear Algebra

First Exam, 9 October 2019

Duration: 90 minutes

- In questions 1-9, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (10 pts)	
Q2 (10 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (10 pts)	
Q7 (10 pts)	
Q8 (15 pts)	
Q9 (15 pts)	
Total	

1. [10 pts.] Show that if B and C are both inverses of the matrix A , then $B = C$.
2. [10 pts.] Show that if A is an $m \times n$ matrix, then AA^T is symmetric.
3. [10 pts.] Let A be a square matrix. Show that if B is a square matrix satisfying $BA = I$, then $B = A^{-1}$.

4. [10 pts.] Find all values of a, b and c , if any, for which $A = \begin{bmatrix} 1 & a & 3 \\ -a & 2 & 15 \\ a+b & 3c+b & 3 \end{bmatrix}$ is symmetric.

5. [10 pts.] Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Suppose C is a 2×1 matrix such that $AC = X$ and $A^{-1}C = B$. Find X .

6. [10 pts.] Let $A = \begin{bmatrix} 1 & 0 & 6 & 0 \\ 2 & 1 & 2 & 3 \\ 3 & 0 & 0 & 1 \\ 1 & 0 & 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 3 & 6 \\ 1 & 0 & 3 & 4 \\ 1 & 0 & 1 & 7 \\ 5 & 0 & 9 & 8 \end{bmatrix}$. Show that AB is singular.

7. [5 + 5 = 10 pts.] Consider the linear system

$$\begin{array}{rrcr} x & +6y & +3z & = 0 \\ 2x & +4y & -2z & = 0 \\ x & -2y & -5z & = 0. \end{array}$$

(a) Determine whether $\begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ is a solution of the above system.

(b) Does the above system have infinitely many solutions? Justify your claim.

8. [10 + 5 = 15 pts.] Let and $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 0 \\ 3 & 0 & 5 \end{bmatrix}$.

(a) Find A^{-1} , if it exists.

(b) Find a matrix C such that $(AC + B)^T = A^T + B$.

9. [5 + 5 + 5 = 15 pts.] Find the general solutions of the linear systems whose augmented matrices are given below.

(a) $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 5 \\ 0 & 1 & 0 & 3 \end{array} \right].$

(b) $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$

(c) $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 1 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right].$

Solutions

1. [10 pts.] Show that if B and C are both inverses of the matrix A , then $B = C$.

Since B and C are both inverses of the matrix A , we have $AB = BA = I$ and $AC = CA = I$. We then have $B = BI = B(AC) = (BA)C = IC = C$, which proves that the inverse of a matrix, if it exists, is unique.

2. [10 pts.] Show that if A is an $m \times n$ matrix, then AA^T is symmetric.

$(AA^T)^T = (A^T)^T A^T = AA^T$. Thus, AA^T is symmetric.

3. [10 pts.] Let A be a square matrix. Show that if B is a square matrix satisfying $BA = I$, then $B = A^{-1}$.

Assume that $BA = I$. To show that A is invertible, it suffices to show that the system $AX = O$ has only the trivial solution. Let X_0 be any solution of this system. If we multiply both sides of $AX_0 = O$ on the left by B , we obtain $BAX_0 = BO$ or $IX_0 = O$ or $X_0 = O$. Thus, the system of equations $AX = O$ has only the trivial solution. That is A is invertible (A^{-1} exists). Now multiply $BA = I$ on both sides by A^{-1} to obtain $BAA^{-1} = IA^{-1}$ or $BI = IA^{-1}$ or $B = A^{-1}$.

4. [10 pts.] Find all values of a, b and c , if any, for which $A = \begin{bmatrix} 1 & a & 3 \\ -a & 2 & 15 \\ a+b & 3c+b & 3 \end{bmatrix}$ is symmetric.

A is symmetric if $a = -a$, $a + b = 3$, and $3c + b = 15$. Thus, $a = 0$, $b = 3$, and $c = 4$.

5. [10 pts.] Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Suppose C is a 2×1 matrix such that $AC = X$ and $A^{-1}C = B$. Find X .

$A^{-1}C = B \Rightarrow C = AB$. Now since $AC = X$, we get

$$X = A(AB) = A^2B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 27 \\ 59 \end{bmatrix}.$$

6. [10 pts.] Let $A = \begin{bmatrix} 1 & 0 & 6 & 0 \\ 2 & 1 & 2 & 3 \\ 3 & 0 & 0 & 1 \\ 1 & 0 & 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 3 & 6 \\ 1 & 0 & 3 & 4 \\ 1 & 0 & 1 & 7 \\ 5 & 0 & 9 & 8 \end{bmatrix}$. Show that AB is singular.

Since the second column of B consists entirely of zeros, then the second column of AB , which is $A\text{col}_2(B)$, consists entirely of zeros. Thus, AB is singular.

7. [5 + 5 = 10 pts.] Consider the linear system

$$\begin{aligned} x + 6y + 3z &= 0 \\ 2x + 4y - 2z &= 0 \\ x - 2y - 5z &= 0. \end{aligned}$$

- (a) Determine whether $\begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ is a solution of the above system.

Since $\begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ satisfies all three equations, it is a solution of the given homogeneous linear system.

(b) Does the above system have infinitely many solutions? Justify your claim.

Since the homogenous system has a nontrivial solution, it has infinitely many solutions.

8. [10 + 5 = 15 pts.] Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 0 \\ 3 & 0 & 5 \end{bmatrix}$.

(a) Find A^{-1} , if it exists.

$$A^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ -3 & 1 & 1 \\ 2 & 0 & -1 \end{bmatrix}$$

(b) Find a matrix C such that $(AC + B)^T = A^T + B$.

Since $(AC + B)^T = (AC)^T + B^T = C^T A^T + B^T = A^T + B$, then $C^T A^T = A^T + B - B^T = A^T$ (B is symmetric). Thus, $C = I_3$.

9. [5 + 5 + 5 = 15 pts.] Find the general solutions of the linear systems whose augmented matrices are given below.

(a) $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 5 \\ 0 & 1 & 0 & 3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$. The general solution is: $x = -1 - t$, $y = 3$, $z = t$,

where $t \in \mathbb{R}$.

(b) $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$.

The general solution is: $x = 7 - 2s - 3t$, $y = s$, $z = t$, where $s, t \in \mathbb{R}$.

(c) $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 1 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$. The system has no solution.